



DEPARTMENT OF  
COMPUTER SCIENCE



# FLOWCHARTS I

**DR JEREMIAH O. BANDELE**

PhD Electrical and Electronic Engineering

University of Nottingham



## LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Explain what flowcharts are and how they are represented.
- Explain various flowchart symbols.
- Give examples of various flowchart symbols.

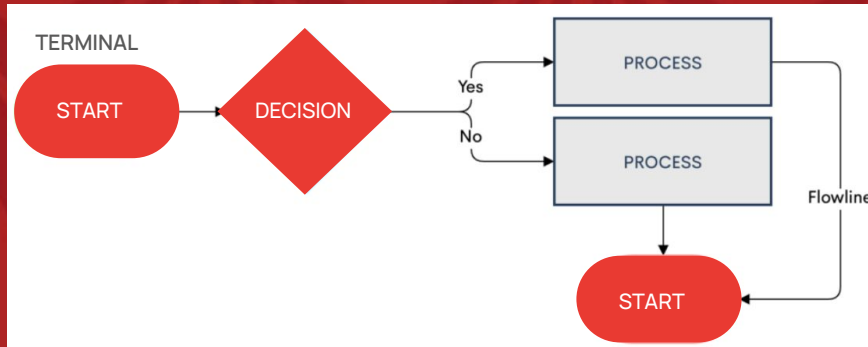


## FLOWCHARTS

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A flowchart is a graphical representation of the sequence of operations in a computer program; or the graphical representation of an algorithm.

Flowcharts use symbols to describe actions such as start/begin, stop/end, decision making and data processing.

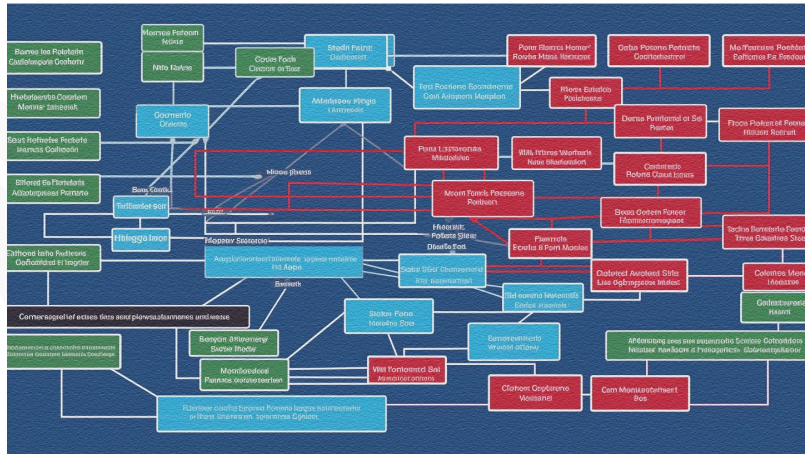




## FLOWCHARTS

With flowcharts, complex processes can be quickly visualised.

A flowchart consists of boxes and lines and the direction of operation is usually from top to bottom and left to right.

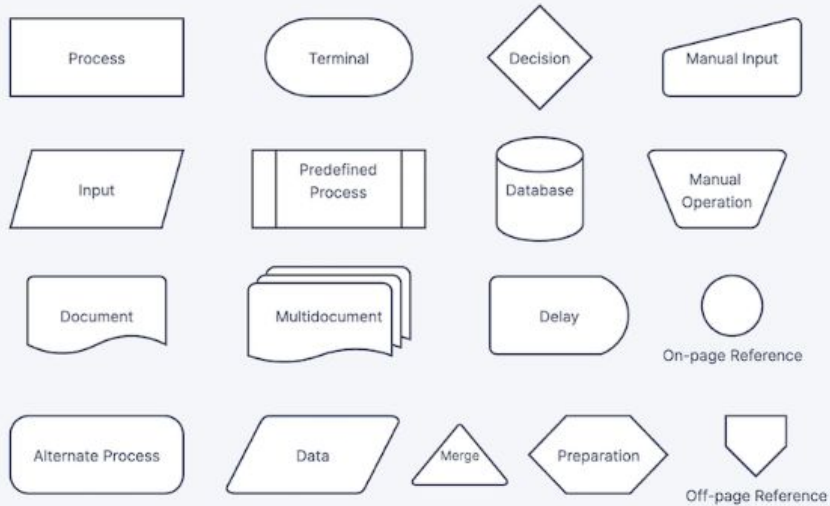




## FLOWCHART SYMBOLS

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A flowchart consists of different symbols that describes different operations.

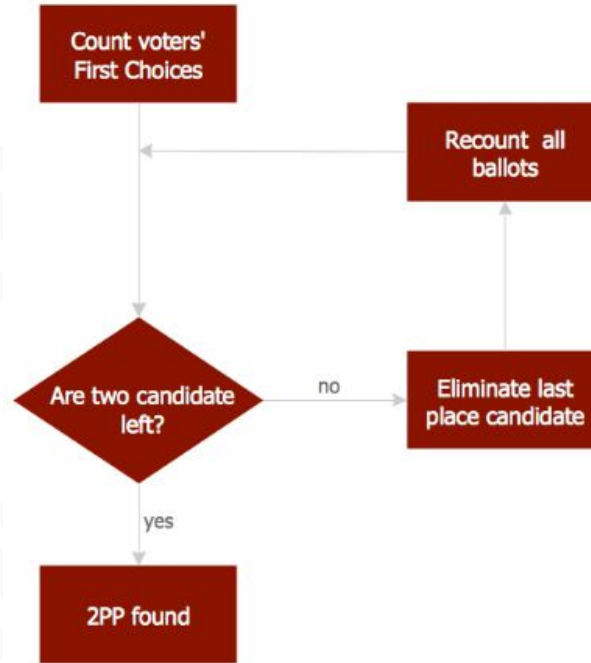




## FLOWCHART SYMBOLS

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The **process** symbol is also called the action symbol and it shows an action, process or function.

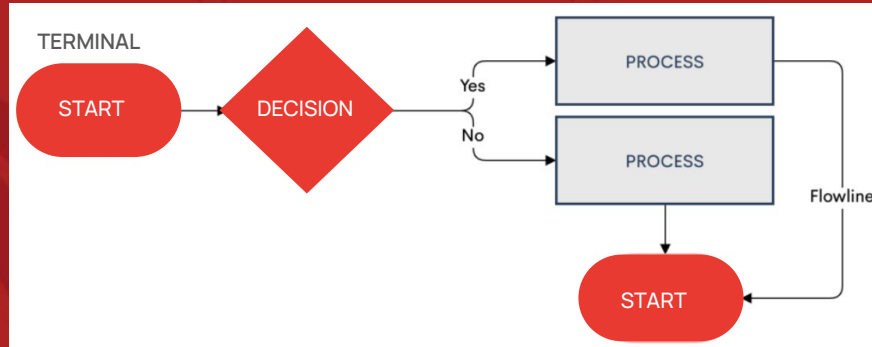






## FLOWCHART SYMBOLS

The start/begin or stop/end symbol is also called a terminator symbol and it shows the start and end points of the flowchart.



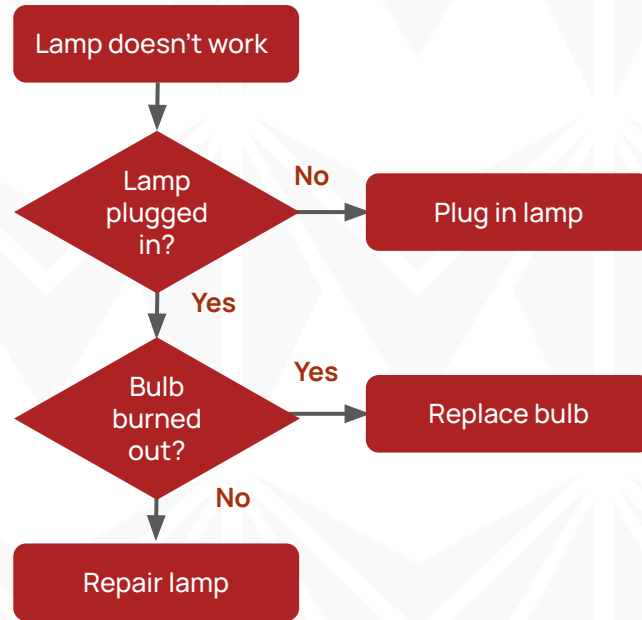


## FLOWCHART SYMBOLS

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The **decision or control** symbol is used when questions such as yes or no and true or false needs an answer.

The answer to these questions will determine the path the algorithm flow will take.

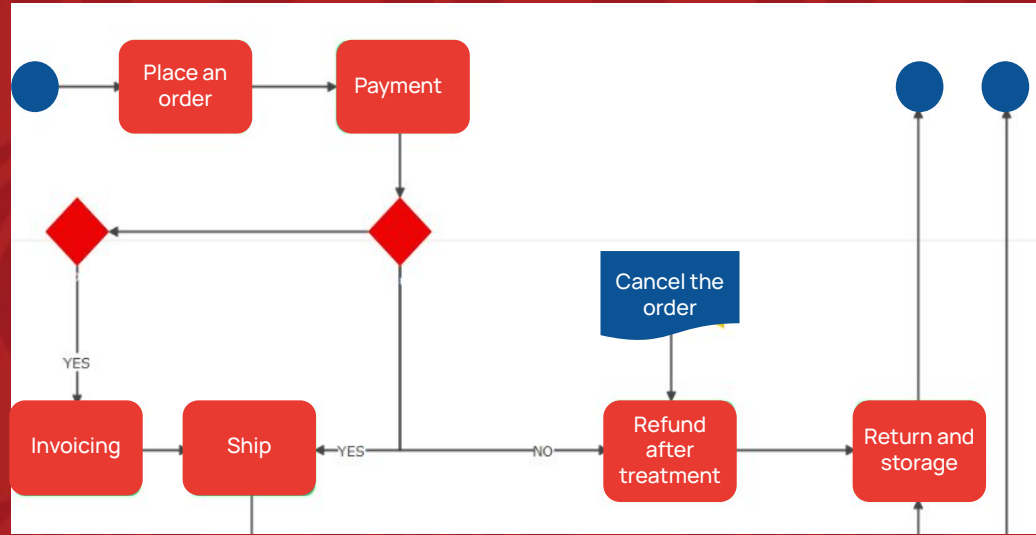






## FLOWCHART SYMBOLS

The document symbol shows a document's input or output.



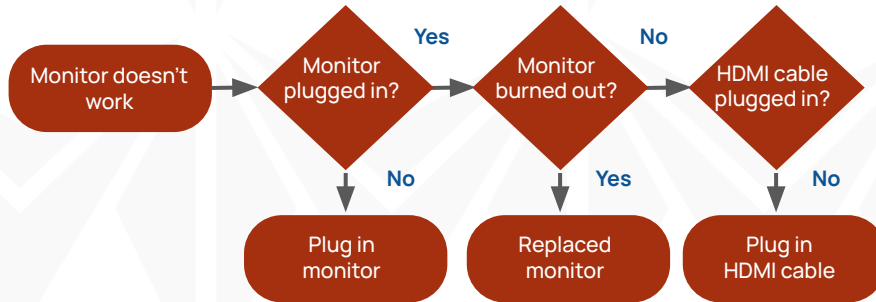


## FLOWCHART SYMBOLS

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The **comment or note** symbol is used to add comments to parts of the flowchart.

We can add comments to the flowchart shown below!

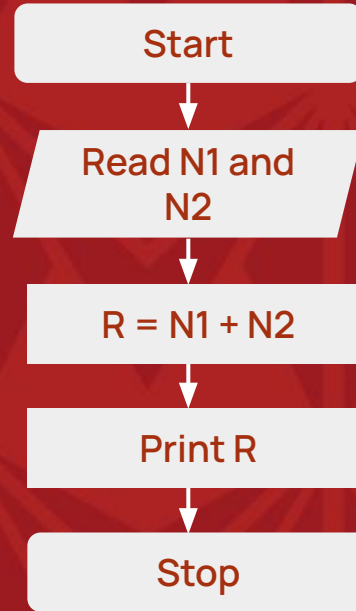




## FLOWCHART SYMBOLS

The **input or output** symbol is also called the data symbol.

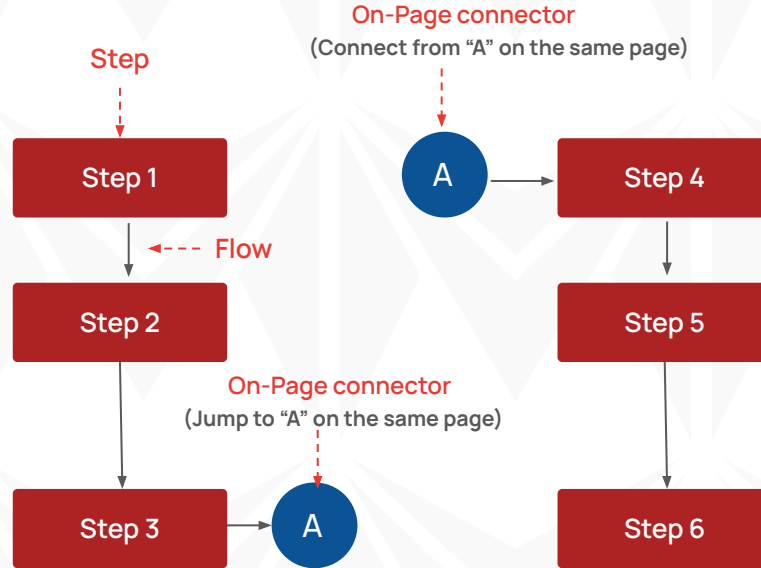
It shows data that can be used either at the input or the output.





## FLOWCHART SYMBOLS

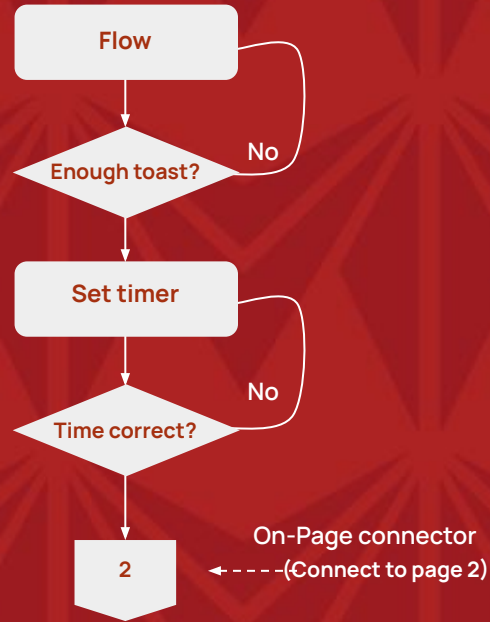
The **on-page connector** symbol is used in complex flowcharts to connect independent elements across a single page.





## FLOWCHART SYMBOLS

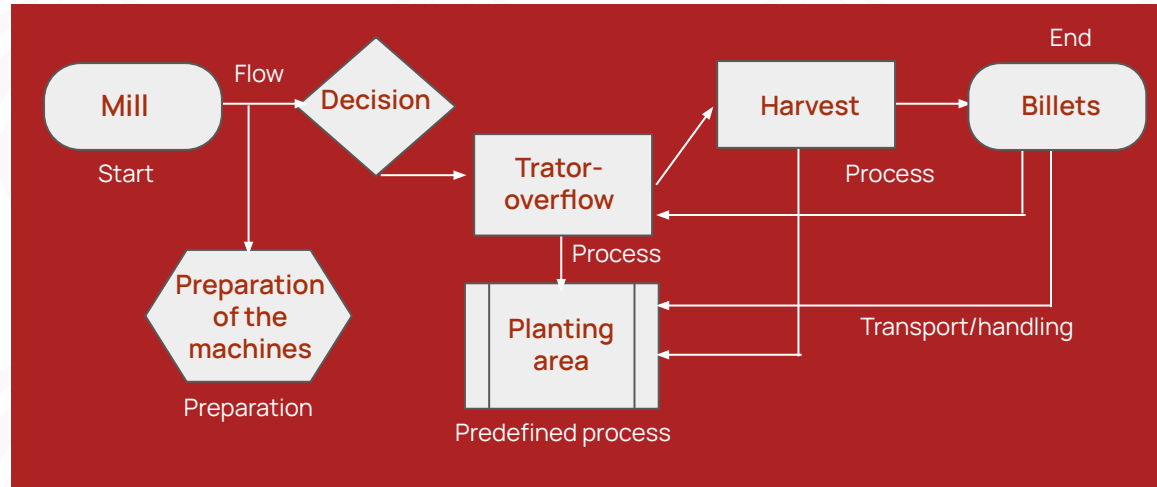
The **link or off-page connector** symbol is used in complex flowcharts to connect independent elements across multiple pages.





## FLOWCHART SYMBOLS

The **predefined process or declaration** symbol is used to show a complex process or a well-known operation.





## MID LESSON QUESTIONS

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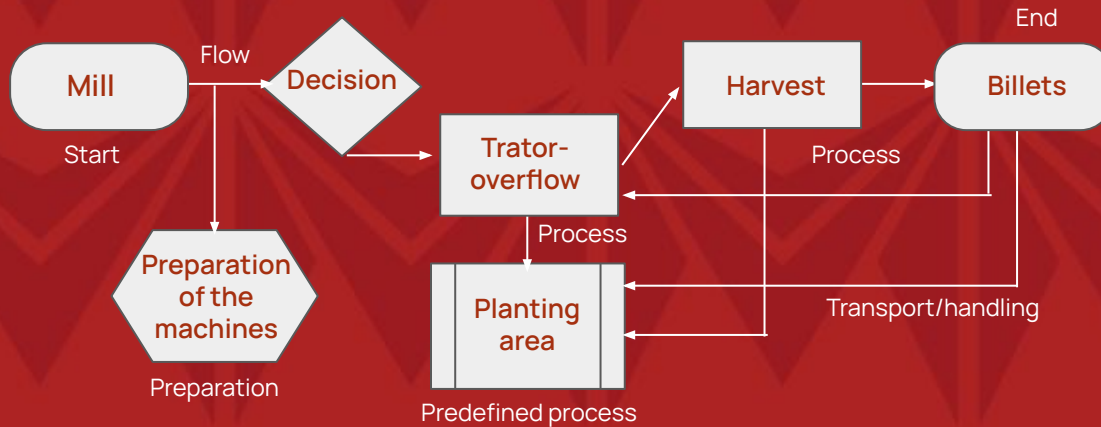
1. The \_\_\_\_\_ symbol is used in complex flowcharts to connect independent elements across multiple pages.
  - a) **link connector**
  - b) on-page connector
  - c) declaration
  - d) flowline





## FLOWCHART SYMBOLS

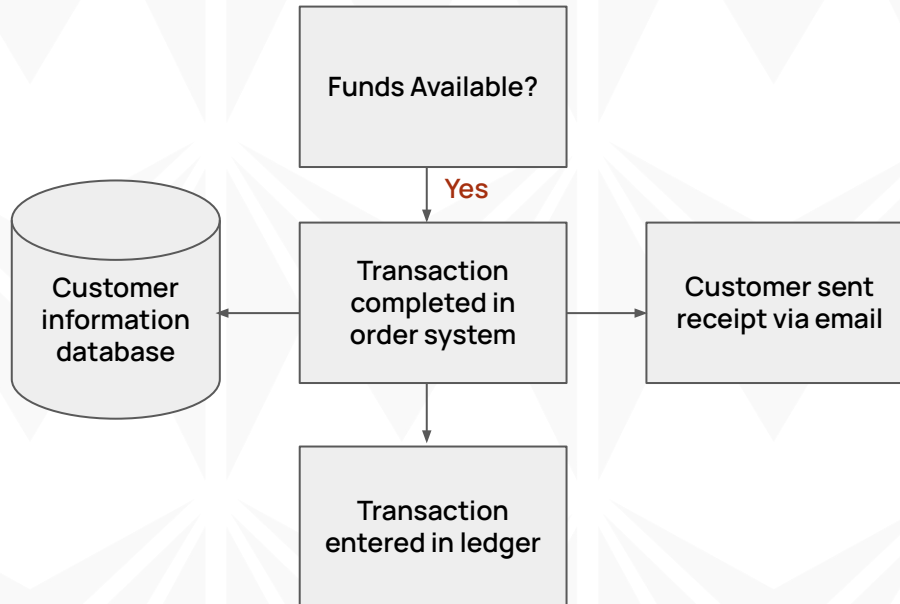
The **preparation** symbol shows the difference between steps preparing for work and those that actually works.





## FLOWCHART SYMBOLS

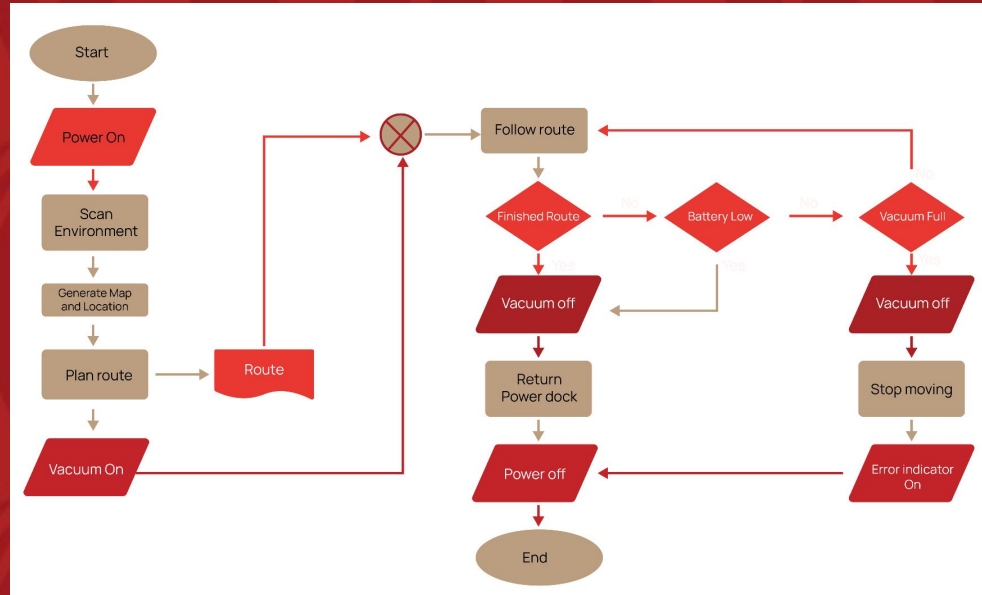
The **database** symbol shows where data is stored.





## FLOWCHART SYMBOLS

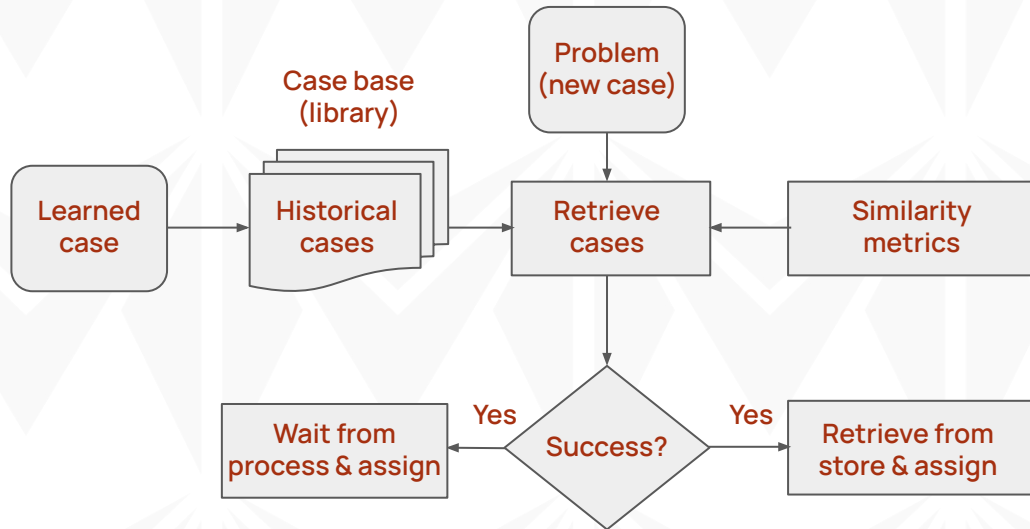
The **summing junction** symbol shows the addition of inputs from various paths.





## FLOWCHART SYMBOLS

The **multiple documents** symbol shows different reports or documents.

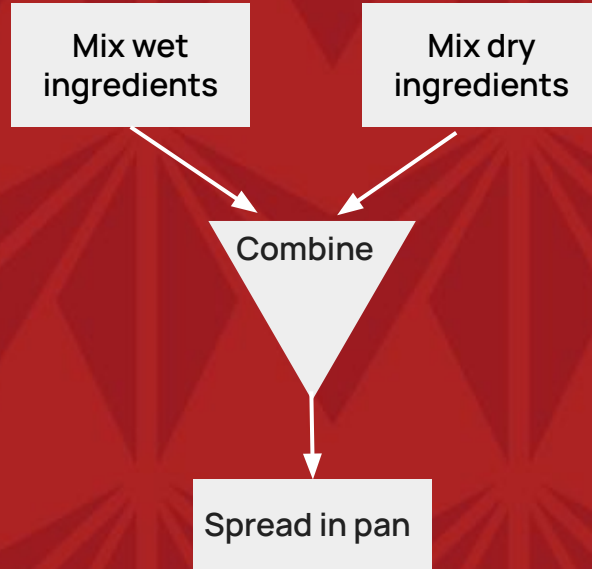




## FLOWCHART SYMBOLS

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The **merge** symbol is used to combine various paths into a single path.





## FURTHER READING RESOURCES

- Chaudhuri, A. B. (2020). Flowchart and algorithm basics: The art of programming. Mercury Learning and Information.
- Chaudhuri, A. B. (2005). The Art of Programming Through Flowcharts & Algorithms. Firewall Media.
- Flowchart Symbol Examples.  
[https://www.google.com/search?q=flowchart+symbols+example&tbm=isch&ved=2ahUKEwjBzdjZvZX\\_AhUn\\_bslHe1pB1MQ2-cCegQIABAA&oq=flowchart+symbols+example](https://www.google.com/search?q=flowchart+symbols+example&tbm=isch&ved=2ahUKEwjBzdjZvZX_AhUn_bslHe1pB1MQ2-cCegQIABAA&oq=flowchart+symbols+example).



## REFERENCES

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- Chaudhuri, A. B. (2020). Flowchart and algorithm basics: The art of programming. Mercury Learning and Information.
- Chaudhuri, A. B. (2005). The Art of Programming Through Flowcharts & Algorithms. Firewall Media.



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# THANK YOU